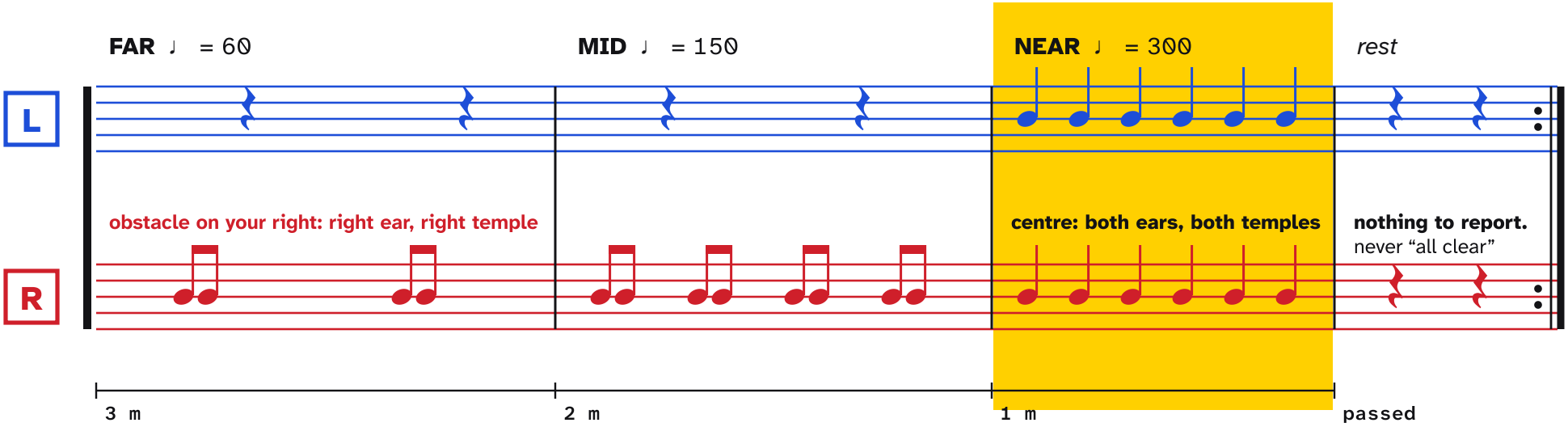
# Determine

**Dumb glasses. Your phone does the thinking. A warning you can hear or feel.**



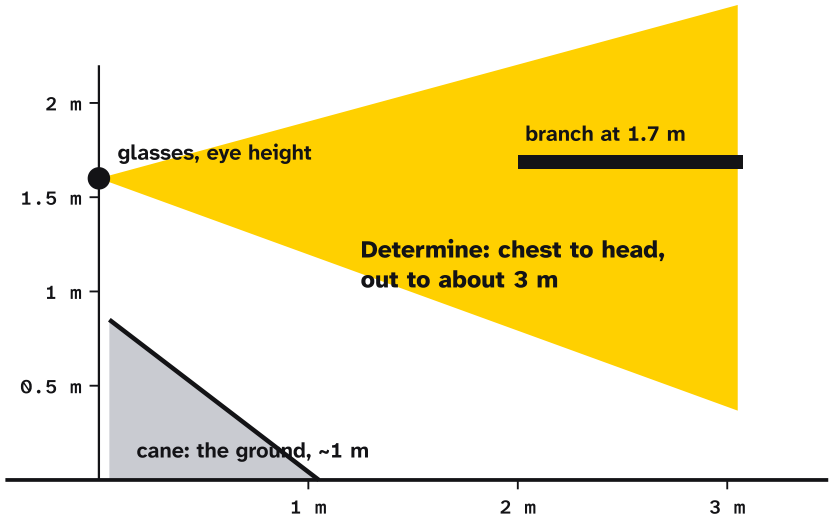

## Cars beep when we can't see behind us. People who are blind deserve a head-level warning too.

**Proposed design · HackTX 26**
Not built or tested yet
UT Austin, 24–25 Oct 2026
For judges, Meta and Annapurna Labs
Full architecture contract on request

# A A cane finds the ground. The branch is at head height.

## About 1 in 8 blind travellers hit their head at least once a month.

In a survey of 307 blind travellers, 13% had a head-level accident at least monthly. 86% of head-level accidents happened outdoors, mostly on tree branches, and 23% had medical consequences.[2] A long cane sweeps the ground ahead of you. Research classes it as a primary aid,[1] and it is excellent at curbs and steps. It cannot feel a branch at eye level, an open cabinet door or a truck mirror.




> **The long cane "does not protect users from obstacles at the upper body and head levels."**
>
> Pittet et al., *Scientific Reports*, 2026[1]

### Maya

ILLUSTRATIVE TRAVELLER

Walks to class with a long cane. Her left ear is weaker than her right.

### Keeps her cane

Determine adds a head-level layer. It never replaces the cane or a guide dog, and it never says the path is clear.

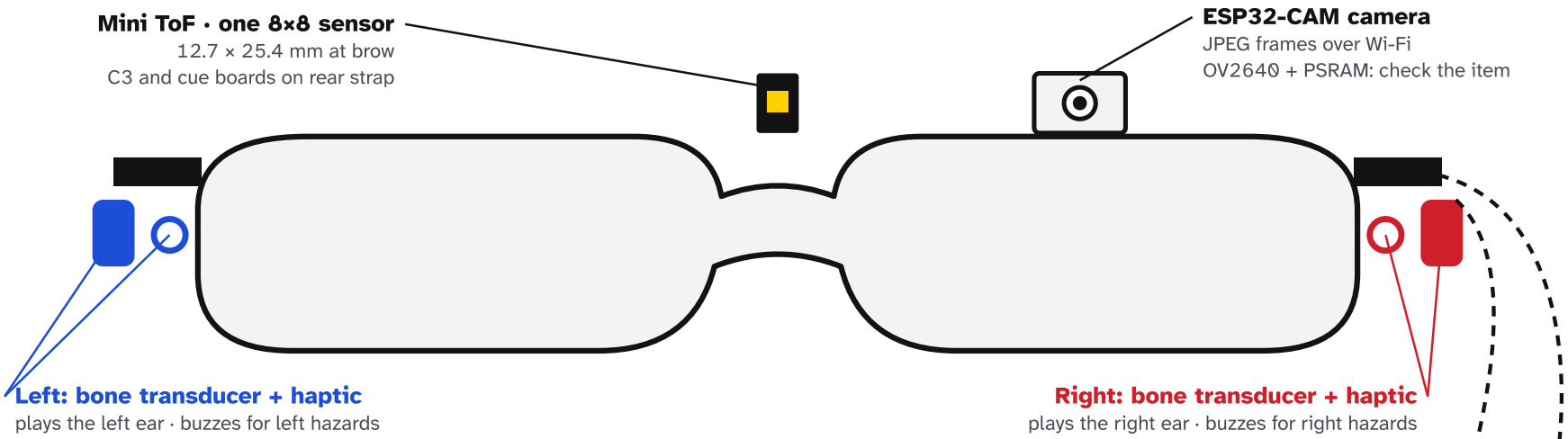
### Wants a warning, not a narrator

In a 2026 trial, every participant preferred short, precise output ("person to your left at 11 o'clock, 3 metres") to generative descriptions.[1]

### Learns it in minutes

4 of 13 trial users struggled with a rival's spatialised sounds.[1] Determine uses a small cue set, taught at setup, with direction carried twice.

# B Dumb glasses. Your phone does the thinking.




Clear or tinted lenses: camera and ToF look out from the front.

power bank in a pocket; board fit is untested

### Sense · small front, boards behind

The ESP32-CAM sends JPEG frames; one compact VL53L5CX Mini sends 64 distances through the rear-mounted C3. Two independent Wi-Fi streams reach the phone, which makes every hazard decision.

### Decide · on the phone

The phone aligns the two streams by timestamp and receive age, then an on-device object detector plus the distance grid feed fixed hazard rules and pick *one* cue: which sound, how loud on each side, which temple buzzes.

### Play · on the glasses

The C3 plays a stored sound straight to the bone over I2S, so urgent beeps skip Bluetooth audio delay. Every command expires: a late one is dropped, never played late.

**Range plus context.** The 8×8 ToF grid measures distance even when a branch or open door has no detector label; the camera adds object context.

**Same app, other glasses.** Meta's Wearables DAT 1.0 can stream supported glasses' camera to a phone; its Mock Device Kit simulates it.[9] Determine's frame input could accept Meta frames. Claim only if built at HackTX.

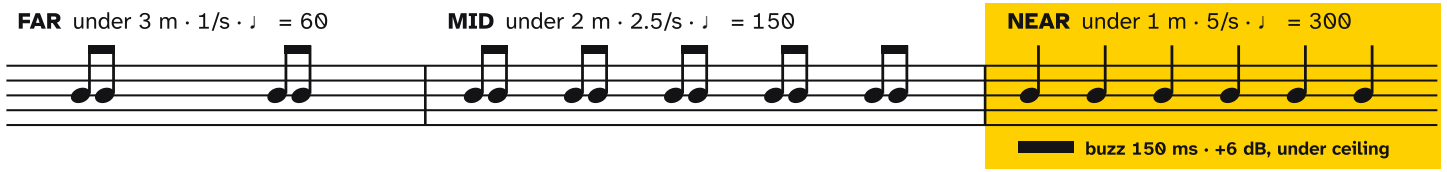
**Two boards, one hotspot.** The C3 has no camera or Espressif parallel-camera driver support,[10] so a separate ESP32-CAM supplies JPEG once its sensor and PSRAM are checked. Both join the phone hotspot; the phone drops stale frames and grids after clock-offset and receive-age checks. No inter-board wire.

**Weight off the nose.** Only camera and 12.7 × 25.4 mm Mini ToF face forward; C3, amps and drivers go on a rear strap, power bank in a pocket.[7] Fit and comfort remain untested. Two separately powered boards share one phone hotspot.

# C A parking sensor that knows what it's beeping at.

### How close → tempo

The car-sensor model everyone already knows. Never above 6 notes a second.[3]



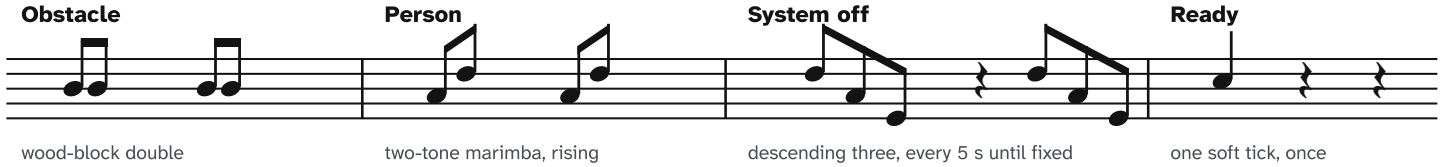

### Where → which ear

Left staff plays the left ear, right staff the right. The matching temple buzzes too.




### What → timbre

Timbre and rhythm are the strongest cues for telling sounds apart.[3]




### Words → optional

Sped-up spoken words (spearcons) after the beep, never instead of it.[4]

**"left" "door" "person" "right"**

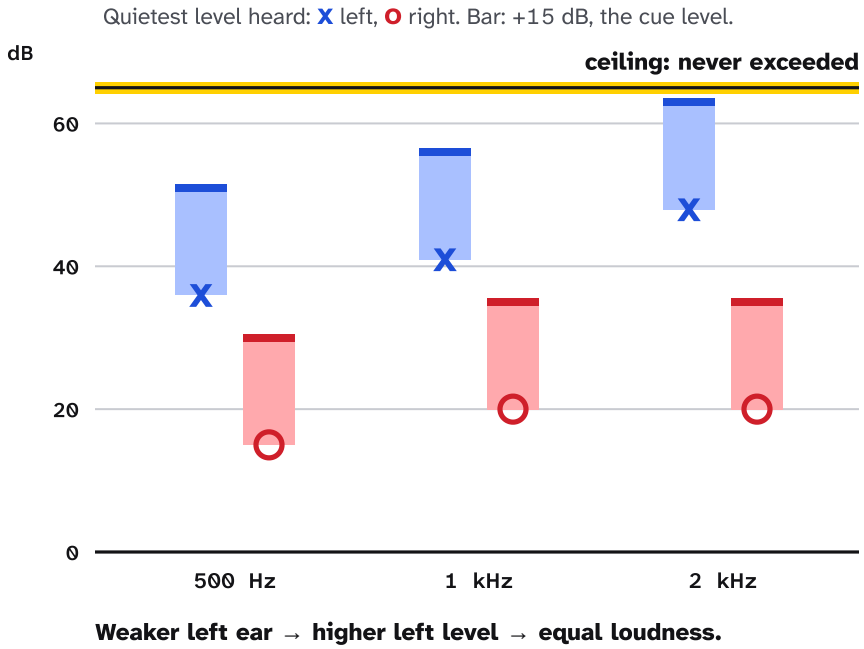
**"11 o'clock, 2 metres"**



**There is no all-clear sound. Silence is a rest, not the end of the piece.**

Silence would mean “nothing to report from a working system”, never “path clear”. If the camera, distance grid or either board's link fails, the design targets a descending **system off** motif within one second, repeated until recovery. If the cue board loses power, the glasses cannot beep: the phone must speak or vibrate the failure instead. Timing and audibility are untested. Tempo, motif and haptics carry urgency too.

# D Equally loud in both ears, even when one is weaker.




**Weaker left ear → higher left level → equal loudness.**

Uses the audiology convention (**X** left, **O** right) but it is not a hearing test and never shows results as one. Values are illustrative.

## Two to three minutes at setup, a 20-second check each time you put them on.

**1 Fit**
Seat the frame. The app notes which output is in use and which side feels weaker.

**2 Find each side's quietest audible level**
An adaptive staircase, tap “heard” or “not heard”, in the three bands the cues use.

**3 Set the level**
10–20 dB above that threshold, your choice, clamped to the ceiling.[3]

**4 Balance**
The real obstacle cue alternates left and right. Nudge until both feel the same.

**5 Verify**
Name the side of five random cues. Four right keeps stereo; otherwise switch mode.

### Stereo

Each ear gets its own calibrated level. Side is heard, and the matching temple confirms it.

### One deaf ear

Bone-conducted sound crosses the skull losing only about 0–10 dB,[5] so left/right audio can't carry direction. Audio goes the same to both sides; the temple buzz and words carry direction.

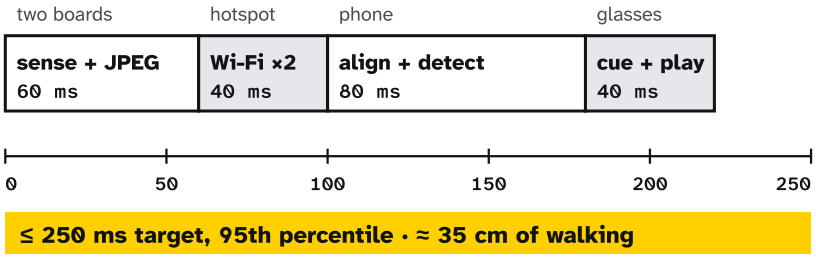
### Haptic only

For anyone who prefers it. No calibration step, slider or urgency boost ever passes the ceiling; if the weak side would need more, it's marked unusable.

# E Two loops. One can't wait. One can think.

Warnings never touch the network. Scene description is a question Maya chooses to ask; the design target is AWS's Annapurna silicon: a Graviton API and a small vision-language model on Inferentia2. Trainium2 is the documented alternative if the Inf2 path fails.[11,13-15]

## Reflex PHONE · OFFLINE · ALWAYS ON




**≤ 250 ms target, 95th percentile · ≈ 35 cm of walking**

Target ≤ 250 ms from capture to cue at the 95th percentile, shown live on the judges' screen. It's a goal until we measure it, and nothing is measured yet — the shared hotspot carrying two streams is the least certain part. At walking pace (1.4 m/s), 250 ms is about 35 cm, so warning distances budget for it.

Three tiers of silicon, placed by latency, power and failure budget: two ESP32-class boards on the glasses (a C3 for grid, cues and haptics[6]; a classic-ESP32 camera[10]), the phone for every decision, and Annapurna silicon for the one job that can wait seconds.

## Describe ON REQUEST · ANNAPURNA SILICON

| | |
|---|---|
| ASK | Long-press: **"what's ahead?"** Never automatic. |
| REDACT | The phone would blur faces and sensitive details before upload; imperfect redaction is a privacy risk, so upload requires the user's choice. |
| GRAVITON | A small arm64 Lambda API (Graviton2) would hold the prompt: two sentences, surroundings first, no guesses about people.[17] |
| INFERENTIA2 | SmolVLM on one inf2.xlarge ($0.76/h) using Hugging Face Optimum Neuron. The model family is listed for Inf2; this deployment is still untested by us.[14,15] |
| SPEAK | The phone would say **"may be wrong"**, then read the result. Any mid- or near-range hazard interrupts it. |

Kept short on purpose: trial participants preferred concise, precise output to generative description.[1] End-to-end time is unmeasured; we plan for seconds, not milliseconds.

### Maya asks

"What's ahead?" The user, not the model, chooses when description is useful. A fast hazard beep would interrupt this answer.

### "May be wrong. A doorway is open on your right; a person is farther ahead near the wall."

ILLUSTRATIVE COPY · NOT MODEL OUTPUT OR A NAVIGATION INSTRUCTION

Describe adds context after the urgent Reflex loop. It never reassures the user that an area is safe, and cloud failure never silences the local cue. Time to response is not measured.

### Inferentia2 · target

SmolVLM via Hugging Face Optimum Neuron on inf2.xlarge. Supported model family, not a validated Determine deployment.[14,15]

### Trainium2 · alternative

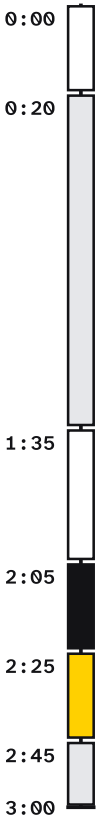
AWS Neuron documents Qwen2-VL-7B on one chip; fit on a smaller trn2.3xlarge is untested.[12,13,16]

### Bedrock · fallback

Nova Lite can handle images, but Bedrock does not guarantee the silicon. New accounts start with Inf/Trn quota 0.[18,19]

Annapurna Labs is HackTX 26's first platinum partner, but no Annapurna challenge is published yet, so Describe stays optional.[21] Request accelerator access now; approval is not guaranteed. Compile and integrate the model at HackTX under its all-work-at-the-event rule.[20]

# F Three minutes, one course, one pulled plug.



**0:00 Hook** 20 s
The parking-sensor line, moved from the bumper to the brow.

**0:20 Live course** 75 s
A spotter walks the wearer past a head-height foam bar, a person, then a left and a right obstacle. The judges' screen shows both streams' age, the 8×8 grid, the chosen cue and measured latency.

**1:35 One-ear mode** 30 s
Switch to a single-sided profile: audio goes mono, the temples carry direction.

**2:05 Pull the plug** 20 s
Unplug the camera board. "System off" plays; reconnect and wait for ready. Never silence.

**2:25 Ask** 20 s
After recovery, "What's ahead?" If cloud is ready, send a redacted frame for a two-sentence description.

**2:45 Close** 15 s
What we measured, what it can't see, what's next.

**3:00**

## The parts, Houston-first

| | |
|---|---|
| C3 · 3-PACK | Micro Center Houston SKU 069005; one used, two spare. User reports stock; price and pickup not independently verified.[6] |
| ESP32-CAM | Micro Center SKU 221739; one used, one spare. Check actual PSRAM and sensor before relying on vision.[10] |
| MINI TOF | One 8×8 VL53L5CX, 12.7 × 25.4 mm. Ship with one Qwiic-to-wire lead; no three-board local cluster.[7] |
| CUES + HAPTICS | Two each: bone transducers, I²S amps, haptic drivers and motors; one I²C mux. Houston Micro Center SKUs in the architecture. Raw 8 Ω bone pair ships from DigiKey.[8] |
| POWER + FIT | One 3.3 V regulator, FTDI programmer, pocket power bank, insulated leads and mounts. Front camera/ToF, boards on rear strap. Build and measure fit. |

Full quantities, Houston SKUs, cables and sources: Architecture §12.2. No grand total or guaranteed delivery: local stock is user-reported, and the Mini ToF and raw bone transducers are shipped exceptions. The prototype is a headband layout, not proven comfortable glasses.

## Built at HackTX, by four people

The official rule: "All work must be done at the event."[20] Parts are bought ahead; firmware, app and frame are built on site.

**Glasses firmware**
Two boards: C3 runs grid, cues, haptics; ESP32-CAM streams JPEG

**Phone perception**
Stream alignment, detector, hazard rules, judge screen

**Sound and calibration**
Cue bank, balance flow, one-ear mode

**Describe and demo**
Graviton API, Neuron model, the pitch

Demo branches: if accelerator access fails, skip the scene answer and show the offline warning loop. If the camera module cannot stream, use the phone camera on a headband as a labelled software fallback — not the proposed glasses. No prerecorded result or fabricated latency; sponsor integration is a goal, not a claim.

# G What it can't see, said out loud.

| | |
|---|---|
| **Glass doors and windows** | LIGHT PASSES THROUGH |
| **Thin wires and ropes** | TOO SMALL FOR 8×8 |
| **Drop-offs, curbs, stairs going down** | THE CANE'S JOB |
| **Anything outside the forward view** | 63° DIAGONAL |
| **Fast-moving vehicles** | NOT A TRAFFIC AID |
| **Glare, darkness, rain and fog** | DEGRADES, ANNOUNCED |
| **Anything beyond about 3 m** | BY DESIGN |

Spoken to every new user during setup. Demos run on a staged indoor course with a sighted spotter.

**Fail loud.**
Design target: announce “system off” within one second through a working output, including the phone if the frame loses power. Never call silence a clear path. Not yet tested.

**Unknown is not clear.**
No detections never produces a reassuring sound.

**The cane stays.**
Determine is a secondary aid, on top of a cane or guide dog.

**Claim only what we measured.**
Every threshold and latency here is a target until tested with blind and low-vision people, with an orientation and mobility specialist.

**Sources (15–24 Oct 2026; Houston stock user-reported)**


1. Pittet et al., secondary travel-aid trial, 2026
2. Manduchi & Kurniawan, mobility accidents, 2011
3. Brewster et al., earcon guidelines, 1995
4. Walker et al., spearcons, 2013
5. Stenfelt, sound crossing the skull, 2012
6. Seeed, XIAO ESP32-C3 wiki: 400 KB SRAM, one I2C; Espressif, I2S
7. SparkFun, VL53L5CX ToF, $25.95
8. Adafruit: bone (direct sold out; DigiKey same part), amp, driver, motor
9. Meta, Wearables DAT and Mock Device Kit
10. Espressif, esp32-camera driver: ESP32/S2/S3; PSRAM above-CIF JPEG
11. About Amazon, Annapurna Labs
12. AWS Neuron, NxD release notes, Trn2+
13. AWS Neuron, Qwen2-VL tutorial
14. Hugging Face, Optimum Neuron models
15. AWS, EC2 Inf2 instances
16. AWS, EC2 Trn2 instances
17. AWS, Lambda arm64 / Graviton2
18. AWS, Bedrock Nova Lite model card
19. AWS, EC2 accelerator quotas
20. Freetail Hackers, HackTX 26 FAQ
21. HackTX, Annapurna Labs platinum partner post